

Data Analysis with R and Python — Quiz 1 — April 7, 2025

NAME:

ROLL NUMBER:

For this quiz, we will use the UK 2024 (provisional) road safety vehicles data available from <https://www.data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-accidents-safety-data>. Use the following code to read the data in R from a copy of the data at the course website.

```
DATALOC <- "https://deepayan.github.io/BSDS/2025-01-DARP/slides/data/rcs2024.csv"
rcs2024 <- read.csv(DATALOC)
```

Each row in the dataset describes a vehicle involved in a collision. Multiple vehicles may be involved in a single collision. The `collision_index` variable is an identifier for the collision a particular vehicle was involved in.

The `sex_of_driver` variable is coded as an integer, with values 1 = Male, 2 = Female, 3 = Not known. Missing values are coded as `-1`.

Question 1: Identify the specific collision that involved the largest number of vehicles (hint: the number of vehicles involved was 25). For this collision, [5]

- How many of the drivers were Male?
- How many of the drivers were Female?
- Ignoring missing values, what was the average age of the Male drivers?
- Ignoring missing values, what was the average age of the Female drivers?

```
rcs2024 <- within(rcs2024,
{
  sex_of_driver[sex_of_driver == -1] <- NA
  age_of_driver[age_of_driver == -1] <- NA
  driver_sex <- factor(sex_of_driver, labels = c("Male", "Female", "Not Known"))
})
coll25 <- names(which(table(rcs2024$collision_index) == 25))
dsub25 <- subset(rcs2024, collision_index == coll25)
with(dsub25, tapply(driver_sex, driver_sex, length))
with(dsub25, tapply(age_of_driver, driver_sex, mean, na.rm = TRUE))
```

Question 2: Obtain the subset of the data for collisions that involved exactly one vehicle.

[10]

- How many such collisions are recorded?
- For what proportion of such collisions is the age of the driver missing?
- For what proportion of such collisions is the sex of the driver Not known?
- Ignoring missing values, what was the average age of the Male drivers in such collisions?
- Ignoring missing values, What was the average age of the Female drivers in such collisions?
- Ignoring missing values, What was the average age of drivers with sex Not Known in such collisions?
- What is the most frequent vehicle type involved in such collisions? The meaning of the integer codes of the `vehicle_type` variable is explained in the attached form.
- What is the *least* frequent vehicle type involved in such collisions? Exclude all vehicle types that were not involved in any recorded accident.

```
vars <- c("vehicle_type", "driver_sex", "age_of_driver")
coll1 <- names(which(table(rcs2024$collision_index) == 1))
dsub1 <- subset(rcs2024, collision_index %in% coll1)[vars]
nrow(dsub1)
with(dsub1, sum(is.na(age_of_driver)) / length(age_of_driver))
with(dsub1, sum(driver_sex == "Not Known") / length(age_of_driver))
with(dsub1, tapply(age_of_driver, driver_sex, mean, na.rm = TRUE))
with(dsub1, table(vehicle_type) |> sort()) ## car / ridden horse
```

Question 3: Using graphical tools you have learned about, study the empirical distribution of `age_of_driver` for different values of `sex_of_driver` (in the entire dataset). Briefly summarize your observations, keeping the following underlying questions in mind: [10]

- Does the age distribution of drivers involved in accidents vary by their sex?
- If we wanted to summarize the age distributions by sex, what would be an appropriate measure (e.g., sample mean, sample median, something else?)

```
library(lattice)
bwplot(driver_sex ~ age_of_driver | I(age_of_driver > 40), data = dsub1)
bwplot(driver_sex ~ age_of_driver, data = dsub1, panel = panel.violin)
stripplot(driver_sex ~ age_of_driver, data = dsub1, jitter = TRUE)
qqmath(~ age_of_driver, subset = age_of_driver > 40,
       data = rcs2024, # dsub1,
       distribution = qexp,
       auto.key = TRUE, groups = driver_sex)
densityplot(~ age_of_driver,
            data = rcs2024, plot.points = FALSE,
            auto.key = TRUE, groups = driver_sex)
histogram(~ age_of_driver | driver_sex,
          data = rcs2024, type = "count")
```

Good luck!